Artificial Intelligence MCQ

1.
The “Father of Artificial Intelligence” is:
Alan Turing
Charles Babbage
John McCarthy
None of the above
Hide
Correct Answer
John McCarthy is also known as the “Father of Artificial Intelligence”.
2.
Blind Search can be used for which of the following situations?
Real-Life Simulation
Small Search Space
Advanced Game Theory
None of the above
Hide
Correct Answer
Blind Search doesn’t contain any domain information such as closeness and hence it is best used for a Small Search Space.
3.
In how many category processes is Artificial Intelligence classified in?
Depends on the input nature
5
2
3
Hide
Correct Answer
AI is classified into 3 categories processes: Sensing, Reasoning, and Acting.
4.
If a machine can change its course of action based on the external environment on its own, the machine is called?
Mobile
Intelligent
Both A and B
None of the above
Hide
Correct Answer
Machines that can change their course of action by making their own decisions are said to be Intelligent.
5.
Which of the following is the common language for Artificial Intelligence?
Python
Java

Lisp
PHP
Hide
Correct Answer
While programming can be done in any language, in today's world Python has become the go-to language for AI and ML-related tasks due to its vast and diverse library functionalities.

6.
Which of the following is not an application of artificial intelligence?
Computer Vision
Natural Language Processing
Database Management System
Digital Assistants
Hide
Correct Answer
Other than Database Management System, all the other options are viable fields of study in Artificial Intelligence.

7.
The proposition symbols in AI are?
true and false
true, false, and null
true
false
Hide
Correct Answer
true and false are the 2 proposition symbols in AI.

8.
Which of the following symbols in AI are logical symbols?
Negation
Conjunction
Implication
All of the above
Hide
Correct Answer
The logical symbols in AI are: Negation, Conjunction, Disjunction, Implication, and Biconditional symbols.

9.
Which of the following are informed search methods?
Best First Search
A* Search
Memory Bound Heuristic Search
All of the above
Hide
Correct Answer
All of the above options are informed search methods used in Artificial Intelligence.

10.

The name of the Artificial Intelligence system developed by Daniel Bobrow was?
BACON
SIMD
STUDENT
None of the above
Hide
Correct Answer
The name of the Artificial Intelligence system developed by Daniel Bobrow was STUDENT developed in the Lisp Language in 1964.
11.
Which of the following is a type of artificial intelligence agent?
Learning AI Agent
Simple Reflex AI Agent
Goal-Based AI Agent
All of the above
Hide
Correct Answer
There are a total of 5 AI agents: Simple Reflex Agent, Model-Based Reflex Agent, Goal-based reflex agent, utility-based agent, learning agent.
12.
How many types of recognition are there in AI?
5
3
2
1
Hide
Correct Answer
There are 3 types of recognition in AI: Biometric Identification, Content-Based Image Retrieval, Handwriting Recognition.
13.
Procedural Domain Knowledge in a rule-based system, is in the form of?
Meta-Rules
Control Rules
Production Rules
None of the above
Hide
Correct Answer
Procedural Domain Knowledge in a rule-based system, is in the form of Production Rules.
14.
What are the different types of Artificial Intelligence approaches?
Strong Approach
Weak Approach
Applied Approach
All of the above
Hide

Correct Answer
All the given options are different types of AI approaches used in different scenarios.
15.
‘The Imitation Game’ was the original name of?
LISP
The Turing Test
The Halting Problem
None of the above
Hide
Correct Answer
The Turing Test was originally called the ‘The Imitation Game’ by its creator.
16.
Which of the following architecture is also known as systolic arrays?
MISD
SISD
SIMD
None of the above
Hide
Correct Answer
MISD architecture is also known as systolic arrays.
17.
Decisions of Victory/Defeat are made in Game trees using which algorithm?
DFS
BFS
Heuristic Search
Min/Max Algorithm
Hide
Correct Answer
The Min/Max algorithm is preferred over other search-based algorithms in Game trees to make win/loss decisions optimally.
18.
The correct ways to solve a problem of state-space search are?
Forward from the initial state
Backward from the goal
Both A and B
None of the above
Hide
Correct Answer
The only 2 ways to solve a problem of state-space search problem are given in options (A) and (B).
19.
Machines running LISP are also called?
AI Workstations
Time-Sharing Terminals
Both A and B
None of the above

Hide
Correct Answer
Machines running LISP are also called AI workstations.
20.
How does an AI agent interacts with its environment?
Using sensors and perceivers
Using only sensors
Using only perceivers
None of the above
Hide
Correct Answer
An AI agent perceives and acts upon the environment using Sensors and Actuators. With Sensors, it senses the surrounding, and with Actuators, it acts on it.
21.
What is the work of Task Environment and Rational Agents?
Problem and Solution
Solution and Problem
Observation and Problem
Observation and Solution
Hide
Correct Answer
Task Environments provides an issue and Rational Agents provides the corresponding solution.
22.
How many types of observing environments are there?
2
3
1
0
Hide
Correct Answer
There are 2 types of observing environments: Fully and Partial.
23.
How many types of quantification are there in AI?
1
2
3
4
Hide
Correct Answer
There are 2 types of quantification in AI: Universal and Existential.
24.
Out of the given options, which of the following algorithms uses the least memory?
DFS
BFS
Both A and B are the same

Cannot be compared
Hide
Correct Answer
DFS uses the lesser amount of memory because at a time the path from the node to a root is only stored in the memory stack unlike BFS which will have to store the whole tree within it.
25.
Machines that try to imitate human intuition while handling vague information lie in the field of AI called?
Functional Logic
Fuzzy Logic
Boolean Logic
Human Logic
Hide
Correct Answer
Fuzzy Logic allows machines to mimic human intuition when provided with information that is vague in nature.
26.
What is meant by a “Complete Algorithm”?
If a solution exists, the algorithm will find it before terminating.
It will find the solution in a finite amount of time.
Both A and B.
None of the above.
Hide
Correct Answer
A complete algorithm is an algorithm that finds a solution to a problem(if it exists) in finite time.
27.
The measure of performance of an AI agent is measured using?
Learning Agent.
Changing Agent.
Both A and B.
None of the above.
Hide
Correct Answer
The performance of an AI agent is improved using the learning agent.
28.
Which of the following are valid Machine Learning algorithms?
Linear Regression
K Means Clustering
Naive Bayes
All of the above
Hide
Correct Answer
All of the above algorithms given above are valid ML algorithms.
29.

What are different machine learning methods?
Memorization
Analogy
Deduction
All of the above
Hide
Correct Answer
The machine learning methods include Memorization, Analogy, and Deduction.
30.
Exploratory Learning is another name for?
Supervised learning
Unsupervised learning
Reinforcement learning
None of the above
Hide
Correct Answer
In unsupervised learning, no external supervision is required so the machine learns on the basis of exploring the datasets themselves.
31.
The things considered in the design of a learning element are?
Components
Feedback
Representation
All of the above
Hide
Correct Answer
The 3 main things to be considered in the design of a learning element are: Components, Feedback, and Representation.
32.
The different types of machine learning are?
Supervised
Unsupervised
Reinforcement
All of the above
Hide
Correct Answer
The 3 types of machine learning are: Supervised, Unsupervised, and Reinforcement Learning.
33.
How is a decision reached upon by a decision tree?
No test
Single Test
Double Test
Multiple sequences of tests
Hide
Correct Answer

A decision tree makes its decision based on the results of a sequence of tests.
34.
A hybrid Bayesian Network consists of?
Discrete Variables
Continuous Variables
Both A and B
None of the above
Hide
Correct Answer
Discrete and Continuous variables both act as numerical inputs to a hybrid Bayesian network.
35.
AI agents are composed of?
Architecture
Program
Both A and B
None of the above
Hide
Correct Answer
AI agents implement mapping for perceptions to actions.
36.
For external action selection, which element is used in the agent?
Perceive
Performance
Actuator
None of the above
Hide
Correct Answer
The performance element is used for external action selection in AI agents.
37.
How is the brightness of a pixel on the screen increased?
Increasing the amount of light directed on the screen
Using mirrors
Using sound waves
None of the above
Hide
Correct Answer
The brightness of a pixel is directly proportional to the amount of light incident on the screen.
38.
The process of making 2 logical expressions look identical is called?
Lifting
Unification
Both A and B
None of the above
Hide

Correct Answer
Unification is the process of making 2 logical expressions look identical.
39.
PEAS is an abbreviation for?
Peace, Environment, Action, Sense
Peer, Environment, Actuators, Sensors
Performance, Environment, Actuators, Sensors
Performance, Environment, Actuators, Sense
Hide
Correct Answer
PEAS is an abbreviation for Performance, Environment, Actuators, Sensors.
40.
Knowledge in AI can be represented as?
Predicate Logic
Propositional Logic
Both A and B
None of the above
Hide
Correct Answer
The representation of knowledge in AI can be done both as Predicate and Propositional Logic.
41.
Which of the following are appropriate levels for a knowledge-based AI agent?
Knowledge Level
Logical Level
Implementation Level
All of the above
Hide
Correct Answer
The 3 main levels for a knowledge-based AI agent are: Knowledge, Logical, and Implementation level.
42.
How can we achieve Artificial Intelligence in real life?
Machine Learning
Deep Learning
Both A and B
None of the above
Hide
Correct Answer
Machine learning and Deep Learning are 2 of the ways to implement Artificial Intelligence in real life.
43.
Which of the following are heuristic search algorithms?
Best First Search Algorithm
A* Search Algorithm
Both A and B

None of the above
Hide
Correct Answer
Both options A and B are heuristics-based search algorithms.
44.
How do we multiply 2 numbers in LISP?
* a b
a * b
a mul b
None of the above
Hide
Correct Answer
The syntax to multiply 2 numbers in LISP is correctly shown in option A.
45.
Inference engines work on the principle of?
Backward Chaining
Forward Chaining
Both A and B
None of the above
Hide
Correct Answer
The modes on which inference engines work are forward and backward chaining.
46.
The process of breaking an image into parts is called?
Smoothing
Segmentation
Processing
Break Down
Hide
Correct Answer
Segmentation is the process of breaking an image into groups based on the similarity of pixels.
47.
Which of the following are valid 3D image processing techniques?
Motion
Texture
Contour
All of the above
Hide
Correct Answer
All of the above options are valid 3D image processing techniques.
48.
Parsing is used for?
Parse tree buildup
Data Interpretation
Sound Recognition

None of the above
Hide
Correct Answer
The process of building up a parse tree for an input string is called parsing.
49.
Artificial Intelligence is associated with computers of which generation?
Second
First
Fifth
Third
Hide
Correct Answer
Fifth Generation computers are said to be able to have Artificial Intelligence within them.
50.
Components of an expert system are?
Knowledge base
User interface
Inference engine
All of the above
Hide
Correct Answer
All the above options comprise the components of an expert system.
51.
Which of the following search method takes less memory space?
Depth First Search
Breadth-First Search
Linear Search
Optimal Search
Hide
Correct Answer
Depth First Search takes the least memory consumption because only the nodes on the current path are stored.
52.
Which of the following is not a type of AI?
Weak AI
Theory of Mind
Reactive Machines
All of the above
Hide
Correct Answer
Reactive Machines are not a type of AI based on functionality.
53.
What is state space in AI?
Collection of all the problem states.
A specific problem states out of all the problem states.
Both A and B.

None of the above
Hide
Correct Answer
State Space in AI is defined as the collection of all the problem states.

Artificial Intelligence MCQ (Multiple Choice Questions)

1. What is the full form of “AI”?
a) Artificially Intelligent
b) Artificial Intelligence
c) Artificially Intelligence
d) Advanced Intelligence
View Answer
Answer: b
Explanation: AI is abbreviated as Artificial Intelligence. It is used to create systems or build machines to think and work like humans.

2. What is Artificial Intelligence?
a) Artificial Intelligence is a field that aims to make humans more intelligent
b) Artificial Intelligence is a field that aims to improve the security
c) Artificial Intelligence is a field that aims to develop intelligent machines
d) Artificial Intelligence is a field that aims to mine the data
View Answer
Answer: c
Explanation: Artificial Intelligence is the development of intelligent systems that work and react in the same way that humans do. Intelligence is a process or a component of the ability to achieve goals in the world. People, animals, and a few machines all have different types and degrees of intelligence.

3. Who is the inventor of Artificial Intelligence?
a) Geoffrey Hinton
b) Andrew Ng
c) John McCarthy
d) Jürgen Schmidhuber
View Answer
4. Which of the following is the branch of Artificial Intelligence?
a) Machine Learning
b) Cyber forensics

c) Full-Stack Developer
d) Network Design
View Answer
Answer: a
Explanation: Machine learning is one of the important sub-areas of Artificial Intelligence likewise Neural Networks, Computer Vision, Robotics, and NLP are also the sub-areas. In machine learning, we build or train ML models to do certain tasks.

5. What is the goal of Artificial Intelligence?
a) To solve artificial problems
b) To extract scientific causes
c) To explain various sorts of intelligence
d) To solve real-world problems
View Answer
Answer: c
Explanation: Artificial Intelligence's goal is to explain various sorts of intelligence.

6. Which of the following is an application of Artificial Intelligence?
a) It helps to exploit vulnerabilities to secure the firm
b) Language understanding and problem-solving (Text analytics and NLP)
c) Easy to create a website
d) It helps to deploy applications on the cloud
View Answer
Answer: b
Explanation: Language understanding and problem-solving come under the NLP and Text Analysis area which involves text recognition and sentiment analysis of the text. NLP ML model is trained to mainly do the task which processes human language's speech or text. For example voice assistant.

7. In how many categories process of Artificial Intelligence is categorized?
a) categorized into 5 categories
b) processes are categorized based on the input provided
c) categorized into 3 categories
d) process is not categorized
View Answer
Answer: c
Explanation: It is categorized into 3 steps Sensing, Reasoning, Acting
i) Sensing: Through the sensor taking in the data about the world
ii) Reasoning: Reasoning is thinking or processing the data sensed by the sensor.
iii) Action: On the basis of input and reasoning, acting is generating and controlling actions in the environment.

8. Based on which of the following parameter Artificial Intelligence is categorized?
a) Based on functionally only
b) Based on capabilities only

c) Based on capabilities and functionally
d) It is not categorized
View Answer
Answer: c
Explanation: The two main categorizations of AI are based on the capability and functionality. Based on capability it is divided into Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Super Intelligence (ASI). Based on functionality it is divided into reactive machines, limited memory, theory of mind, and self-awareness.

9. Which of the following is a component of Artificial Intelligence?
a) Learning
b) Training
c) Designing
d) Puzzling
View Answer
Answer: a
Explanation: Intelligence is intangible and is composed of mainly five techniques. Learning is the process of gaining knowledge by understanding, practicing, being taught, or experiencing one thing. Learning enhances the awareness of any topic, hence learning is one of the important components.

10. What is the function of an Artificial Intelligence "Agent"?
a) Mapping of goal sequence to an action
b) Work without the direct interference of the people
c) Mapping of precept sequence to an action
d) Mapping of environment sequence to an action
View Answer
Answer: c
Explanation: A math function that converts a collection of perceptions into actions is known as the agent function. The function is implemented using agent software. An agent is responsible for the actions performed by the machine once it senses the environment.

11. Which of the following is not a type of Artificial Intelligence agent?
a) Learning AI agent
b) Goal-based AI agent
c) Simple reflex AI agent
d) Unity-based AI agent
View Answer
Answer: d
Explanation: There are mainly 5 types of agents:

12. Which of the following is not the commonly used programming language for Artificial Intelligence?
a) Perl
b) Java

c) PROLOG
d) LISP
View Answer
Answer: a
Explanation: Perl is a scripting language. Whereas other programming languages are used to program AI machines.

13. What is the name of the Artificial Intelligence system developed by Daniel Bobrow?
a) program known as BACON
b) system known as STUDENT
c) program known as SHRDLU
d) system known as SIMD
View Answer
Answer: b
Explanation: STUDENT is the name of the Artificial Intelligence system developed by Daniel Bobrow in 1964. Daniel Bobrow had used LISP programming language to write this AI program for his PhD thesis.

14. What is the function of the system Student?
a) program that can read algebra word problems only
b) system which can solve algebra word problems but not read
c) system which can read and solve algebra word problems
d) None of the mentioned
View Answer
Answer: c
Explanation: The system STUDENT developed by Daniel Bobrow was written in LISP to read and solve algebra word problems of high school books. This is referred as the achievement in the field of Natural Language Processing.

15. Which of the following is not an application of artificial intelligence?
a) Face recognition system
b) Chatbots
c) LIDAR
d) DBMS
View Answer
Answer: d
Explanation: Face recognition system, Chatbots, and LIDAR are the various applications of AI in various fields like security system, business, automobiles etc. DBMS is used to store and manipulate data.

16. Which of the following machine requires input from the humans but can interpret the outputs themselves?
a) Actuators
b) Sensor
c) Agents
d) AI system
View Answer

Answer: d
Explanation: Actuators are used in machines to convert energy from one form to another to perform a physical function. The sensor is a device that receives signals from the physical environment to detect the changes. Systems receive input from humans and interpret the outputs.

17. _________ number of informed search method are there in Artificial Intelligence.
a) 4
b) 3
c) 2
d) 1
View Answer
Answer: a
Explanation: There are four types of informed search methods. The four types of informed search method are best-first search, Greedy best-first search, A* search and memory bounded heuristic search.

18. The total number of proposition symbols in AI are ________
a) 3 proposition symbols
b) 1 proposition symbols
c) 2 proposition symbols
d) No proposition symbols
View Answer
Answer: c
Explanation: There are totally 2 proposition symbols. The two proposition symbols are true and false.

19. The total number of logical symbols in AI are ____________
a) There are 3 logical symbols
b) There are 5 logical symbols
c) Number of logical symbols are based on the input
d) Logical symbols are not used
View Answer
Answer: b
Explanation: There are totally five logical symbols. The five logical symbols are:
a) Negation
b) Conjunction
c) Disjunction
d) Implication
e) Biconditional

20. Which of the following are the approaches to Artificial Intelligence?
a) Applied approach
b) Strong approach
c) Weak approach
d) All of the mentioned
View Answer

Answer: d
Explanation: Strong AI is used to build machines that can truly reason and solve problems. Weak AI deals with building computer-based Artificial Intelligence that can act as if it were intelligent but cannot truly reason and solve problems. Applied approach creates commercially viable “smart” systems.
In the Cognitive approach, a computer is used to test theories about how the human mind works.

21. Face Recognition system is based on which type of approach?
a) Weak AI approach
b) Applied AI approach
c) Cognitive AI approach
d) Strong AI approach
View Answer
Answer: b
Explanation: Applied approach aims to produce commercially viable “smart” systems such as, for example, a security system that recognizes the faces of people to provide access. The applied approach has already enjoyed considerable success.

22. Which of the following is an advantage of artificial intelligence?
a) Reduces the time taken to solve the problem
b) Helps in providing security
c) Have the ability to think hence makes the work easier
d) All of the above
View Answer
Answer: d
Explanation: Artificial intelligence creates a machine that can think and make decisions without human involvement.

23. Which of the following can improve the performance of an AI agent?
a) Perceiving
b) Learning
c) Observing
d) All of the mentioned
View Answer
Answer: b
Explanation: An AI agent learns from previous states by saving it and responding to the same situation better if it occurs again in the future. Hence, learning can improve the performance of an AI agent.

24. Which of the following is/are the composition for AI agents?
a) Program only
b) Architecture only
c) Both Program and Architecture
d) None of the mentioned
View Answer
Answer: c

Explanation: An AI agent program will implement function mapping percepts to actions.

25. On which of the following approach A basic line following robot is based?
a) Applied approach
b) Weak approach
c) Strong approach
d) Cognitive approach
View Answer
Answer: b
Explanation: Weak approach is concerned with the development of a computer-based artificial intelligence that can behave intelligently but cannot really reason or solve issues. According to Weak approach, properly configured computers can mimic human intellect.

26. Artificial Intelligence has evolved extremely in all the fields except for ________
a) Web mining
b) Construction of plans in real time dynamic systems
c) Understanding natural language robustly
d) All of the mentioned
View Answer
Answer: d
Explanation: Artificial Intelligence is used in all the fields to make work easier and complete the work before the deadline. However, it could not excel in these fields. Hence, these areas need more focus for improvements.

27. Which of the following is an example of artificial intelligent agent/agents?
a) Autonomous Spacecraft
b) Human
c) Robot
d) All of the mentioned
View Answer
Answer: d
Explanation: Humans can be considered agents. Sensors include eyes, ears, skin, taste buds, and so on, whereas effectors include hands, fingers, legs, and mouth. Agents are robots. Sensors on robots might include a camera, sonar, infrared, bumper, and so on. Actuators can include grippers, wheels, lights, speakers, and other components. Based on its senses, autonomous spacecraft makes decisions on its own.

28. Which of the following is an expansion of Artificial Intelligence application?
a) Game Playing
b) Planning and Scheduling
c) Diagnosis
d) All of the mentioned
View Answer
Answer: d
Explanation: In recent days AI is used in all sectors in different forms. All sectors require intelligence and automation for its working.

29. What is an AI ‘agent’?
a) Takes input from the surroundings and uses its intelligence and performs the desired operations
b) An embedded program controlling line following robot
c) Perceives its environment through sensors and acting upon that environment through actuators
d) All of the mentioned
View Answer
Answer: d
Explanation: An AI agent is defined as anything that uses sensors and actuators to perceive and act on the environment. It receives information from its surroundings via sensors, executes operations, and outputs via actuators.

30. Which of the following environment is strategic?
a) Rational
b) Deterministic
c) Partial
d) Stochastic
View Answer
Answer: b
Explanation: In a deterministic environment the output is determined based on a particular state. If the environment is deterministic except for the action of other agents it is called deterministic.

31. What is the name of Artificial Intelligence which allows machines to handle vague information with a deftness that mimics human intuition?
a) Human intelligence
b) Boolean logic
c) Functional logic
d) Fuzzy logic
View Answer
Answer: d
Explanation: Many popular goods, such as microwave ovens, cars, and plug-in circuit boards for desktop PCs, employ the first widely-used commercial form of Artificial Intelligence. It enables robots to handle ambiguous data with a dexterity that resembles human intuition.

32. Which of the following produces hypotheses that are easy to read for humans?
a) Machine Learning
b) ILP
c) First-order logic
d) Propositional logic
View Answer
Answer: b
Explanation: ILP (Inductive logic programming) is a subfield of artificial intelligence. Because ILP can participate in the scientific cycle of experimentation So that it can produce a flexible structure.

33. What does the Bayesian network provide?
a) Partial description of the domain
b) Complete description of the problem
c) Complete description of the domain
d) None of the mentioned
View Answer
Answer: c
Explanation: A Bayesian network provides a complete description of the domain.

33. What is the total number of quantification available in artificial intelligence?
a) 4
b) 3
c) 1
d) 2
View Answer
Answer: d
Explanation: There are two types of quantification. They are:
a) Universal
b) Existential

34. What is Weak AI?
a) the study of mental faculties using mental models implemented on a computer
b) the embodiment of human intellectual capabilities within a computer
c) a set of computer programs that produce output that would be considered to reflect intelligence if it were generated by humans
d) all of the mentioned
View Answer
Answer: a
Explanation: Weak AI is the study of mental faculties using mental models implemented on a computer.

35. Which of the following are the 5 big ideas of AI?
a) Perception
b) Human-AI Interaction
c) Societal Impact
d) All of the above
View Answer
Answer: d
Explanation: The 5 big ideas are:

Artificial Intelligence Multiple Choice Questions

1) Artificial Intelligence is about_____.

a. Playing a game on Computer
b. Making a machine Intelligent
c. Programming on Machine with your Own Intelligence
d. Putting your intelligence in Machine

Hide Answer Workspace

Answer: b. Making a machine Intelligent.

Explanation: Artificial Intelligence is a branch of Computer science, which aims to create intelligent machines so that machine can think intelligently in the same manner as a human does.

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2) Who is known as the -Father of AI"?

a. Fisher Ada
b. Alan Turing
c. John McCarthy
d. Allen Newell

Hide Answer Workspace

Answer: c. John McCarthy

Explanation: John McCarthy was a pioneer in the AI field and known as the father of Artificial intelligence. He was not only the known as the father of AI but also invented the term Artificial Intelligence.

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3) Select the most appropriate situation for that a blind search can be used.

a. Real-life situation
b. Small Search Space
c. Complex game
d. All of the above

Hide Answer Workspace

Answer: b. Small Search Space

Explanation: Blind Search is also known as uninformed search, and it does not contain any domain information such as closeness, location of the goal, etc. Hence the most appropriate situation that can be used for the blind search is Small-search Space.

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4) The application/applications of Artificial Intelligence is/are

a. Expert Systems
b. Gaming
c. Vision Systems
d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: All the given options are the applications of AI.

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5) Among the given options, which search algorithm requires less memory?

a. Optimal Search
b. Depth First Search
c. Breadth-First Search
d. Linear Search

Hide Answer Workspace

Answer: b. Depth First Search

Explanation: The Depth Search Algorithm or DFS requires very little memory as it only stores the stack of nodes from the root node to the current node.

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6) If a robot is able to change its own trajectory as per the external conditions, then the robot is considered as the__

a. Mobile
b. Non-Servo
c. Open Loop
d. Intelligent

Hide Answer Workspace

Answer: d. Intelligent

Explanation: If a robot is able to change its own trajectory as per the external conditions, then the robot is considered intelligent. Such type of agents come under the category of AI agents or Rational Agents.

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7) Which of the given language is not commonly used for AI?

a. LISP
b. PROLOG
c. Python
d. Perl

Hide Answer Workspace

Answer: d. Perl

Explanation: Among the given languages, Perl is not commonly used for AI. LISP and PROLOG are the two languages that have been broadly used for AI innovation, and the most preferred language is Python for AI and Machine learning.

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8) A technique that was developed to determine whether a machine could or could not demonstrate the artificial intelligence known as the___

a. Boolean Algebra
b. Turing Test
c. Logarithm
d. Algorithm

Hide Answer Workspace

Answer: b. Turing Test

Explanation: In the year **1950**, mathematician and computing pioneer **Alan Turing** introduced a test to determine whether a machine can think like a human or not, which means it can demonstrate intelligence, known as the **Turing Test**

. It was based on the **"Imitation game"** with some modifications. This technique is still a measure of various successful AI projects, with some updates.

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9) The component of an Expert system is_________.

a. Knowledge Base
b. Inference Engine
c. User Interface
d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: Expert system is a part of AI and a computer program that is used to solve complex problems, and to give the decision-making ability like human. It does this with the help of **a Knowledge base, Inference engine, and User interface**, and all these are the components of an Expert System.

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10) Which algorithm is used in the Game tree to make decisions of Win/Lose?

a. Heuristic Search Algorithm
b. DFS/BFS algorithm
c. Greedy Search Algorithm
d. Min/Max algorithm

Hide Answer Workspace

Answer: d. Min/Max Algorithm

Explanation: A game tree is a directed graph whose nodes represent the positions in Game and edges represent the moves. To make any decision, the game tree uses the Min/Max algorithm. The Min/Max algorithm

is the preferred one over other search algorithms, as it provides the best move to the player, assuming that the opponent is also playing Optimally.

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11) The available ways to solve a problem of state-space-search.

a. 1
b. 2
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12) Among the given options, which is not the required property of Knowledge representation?

a. Inferential Efficiency
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Explanation: Knowledge representation is the part of Artificial Intelligence that deals with AI agent thinking and how their thinking affects the intelligent behavior of agents. A good knowledge representation requires the following properties:

- **Representational Accuracy**
- **Inferential Adequacy**
- **Inferential Efficiency**
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13) An AI agent perceives and acts upon the environment using___.

a. Sensors

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d. Both a and c

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Explanation: An AI agent perceives and acts upon the environment using Sensors and Actuators. With Sensors, it senses the surrounding, and with Actuators, it acts on it.

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b. Simple &Condition-action rule

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Explanation: The simple reflex agent takes decisions only on the current condition and acts accordingly; it ignores the rest of history; hence it follows the Condition-action rule.

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17) Which term describes the common-sense of the judgmental part of problem-solving?

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Explanation: In problem-solving, the Heuristic describes the common sense or Judgemental part.

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b. Six states
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b. Connectives
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Explanation: Complex sentences are built by combining the atomic sentences using connectives.

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26) If according to the hypothesis, the result should be positive, but in fact it is negative, then it is known as_______.

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Explanation: The False Positive Hypothesis means that according to results, you have that condition, but in reality, you don't have it. Such as for a medical test, if someone is found Positive for a disease, but actually he doesn't have that disease, then it comes under the False Positive hypothesis.

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27) A hybrid Bayesian Network consist_____.

a. Discrete variables only

b. Discontinuous Variable

c. Both Discrete and Continuous variables

d. Continuous Variable only

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Answer: c. Both Discrete and Continuous Variables

Explanation: The Hybrid Bayesian network contains both discrete and continuous variables as the numerical inputs. To define the hybrid network, both kinds of distributions are used at wide probability distribution.

---

28) The process of capturing the inference process as Single Inference Rule is known as:

a. Clauses

b. Ponens

c. Generalized Modus Ponens

d. Variables

Hide Answer Workspace

Answer: c. Generalized Modus Ponens

Explanation: For all inference process in FOL, the single inference rule can be used, which is called Generalized Modus Ponens. It is said to be the lifted version of Modus ponens.

Generalized Modus Ponens can be said as, " **P implies Q and P is asserted to be true, therefore Q must be True."**

---

29) Which process makes two different Logical expressions look identical?

a. Unification
b. Lifting
c. Inference Process
d. None of the above

Hide Answer Workspace

Answer: a. Unification

Explanation: Unification is the process of making two different logical expressions identical by finding a substitution.

---

30) Which algorithm takes two sentences as input and returns a Unifier?

a. Inference
b. Hill-Climbing
c. Unify algorithm
d. Depth-first search

Hide Answer Workspace

Answer: c. Unify Algorithm

Explanation: The unify algorithm takes two atomic sentences and return a unifier. It is used for the unification process.

---

31) The PEAS in the task environment is about___________.

a. Peer, Environment, Actuators, Sense
b. Performance, Environment, Actuators, Sensors
c. Perceiving, Environment, Actuators, Sensors
d. None of the above

Hide Answer Workspace

Answer: b. Performance, Environment, Actuators, Sensors

Explanation: PEAS is a representation model on which an AI agent works. It is made up of four words:

- P: Performance
- E: Environment
- A: Actuators
- S: Sensors

---

32) In state-space, the set of actions for a given problem is expressed by the_____.

a. Intermediate States
b. Successor function that takes current action and returns next state
c. Initial States
d. None of the above

Hide Answer Workspace

Answer: b. Successor function that takes current action and returns next state

Explanation: The successor function provides a description of all possible actions and their next states, which means their outcomes.

---

33) In which search problem, to find the shortest path, each city must be visited once only?

a. Map coloring Problem
b. Depth-first search traversal on a given map represented as a graph

c. Finding the shortest path between a source and a destination

d. Travelling Salesman problem

Hide Answer Workspace

Answer: d. Travelling Salesman problem

Explanation: The TSP or Travelling Salesman problem is about finding the shortest possible route to visit each city only once and returning to the origin city when the list of all cities and distances between each pair of cities is given.

---

34) In the TSP problem of n cities, the time taken for traversing all cities, without having prior knowledge of the length of the minimum tour will be_______.

a. O(n)

b. O(n2)

c. O(n!)

d. O(n/2)

Hide Answer Workspace

Answer: c. O(n!)

Explanation: In the TSP problem of n cities, the time taken for traversing all cities without having prior knowledge of the length of the minimum tour will be O(n!).

---

35) Web Crawler is an example of______.

a. Intelligent Agent

b. Problem-solving agent

c. Simple reflex agent

d. Model-based agent

Hide Answer Workspace

Answer: Intelligent Agent

Explanation: The web crawler is an example of Intelligent agents, which is responsible for collecting resources from the Web, such as HTML documents, images, text files, etc.

---

36) The main function of problem-solving agent is to________.

a. Solve the given problem and reach the goal
b. Find out which sequence of action will get it to the goal state.
c. Both a & b
d. None of the above

Hide Answer Workspace

Answer: Both a & b

Explanation: Problem-solving agents are the goal-based agents that use different search strategies and algorithms to solve a given problem.

---

37) In artificial Intelligence, knowledge can be represented as_______.

i. Predicate Logic

ii. Propositional Logic

iii. Compound Logic

iv. Machine Logic

a. Both I and II
b. Only II
c. Both II and III
d. Only IV

Hide Answer Workspace

Answer: a. Both I and II

Explanation: There are several techniques of Knowledge representation in AI, and among them, one is Logical Representation. The logical representation can be done in two ways **Predicate Logic and Propositional Logic**, hence knowledge can be represented as both predicate and Propositional logic.

---

38) For propositional Logic, which statement is false?

a. The sentences of Propositional logic can have answers other than True or False.
b. Each sentence is a declarative sentence.
c. Propositional logic is a knowledge representation technique in AI.
d. None of the above

Hide Answer Workspace

Answer: a. The sentences of Propositional logic can have answers other than True or False

Explanation: Propositional Knowledge or PL is the simplest form of logic that is used to represent the knowledge, where all the sentences are propositions. In this, each sentence is a declarative sentence that can only be either true or False.

Such as, **It is Sunday today.** This sentence can be either true or false only.

---

39) First order logic Statements contains______.

a. Predicate and Preposition
b. Subject and an Object
c. Predicate and Subject
d. None of the above

Hide Answer Workspace

Answer: c. Predicate and Subject

Explanation: The first-order logic

is also known as the First-order predicate logic, which is another way of knowledge representation. The FOL statements contain two parts that are s**ubject and Predicate**.

For e.g., **X is an Integer; In this, X is Subject and Is an Integer is Predicate**.

---

40) A knowledge-based agent can be defined with _____ levels.

a. 2 Levels
b. 3 Levels
c. 4 Levels
d. None of the above

Hide Answer Workspace

Answer: b. 3 Levels

Explanation: The knowledge-based agents have the capability of making decisions and reasoning to act efficiently. It can be viewed at three different levels, which are:

- **Knowledge Level**
- **Logical Level**
- **Implementation Level**

---

41) Ways to achieve AI in real-life are_________.

a. Machine Learning
b. Deep Learning
c. Both a & b
d. None of the above

Hide Answer Workspace

Answer: c. Both a &b

Explanation: Machine Learning and Deep Learning are the two ways to achieve AI in real life.

42) The main tasks of an AI agent are_______.

a. Input and Output

b. Moment and Humanly Actions

c. Perceiving, thinking, and acting on the environment

d. None of the above

Hide Answer Workspace

Answer: c. Perceiving, thinking, and acting on the environment

Explanation: The AI agent is the rational agent that runs in the cycle of Perceive, think, and act.

---

43) The probabilistic reasoning depends upon____________.

a. Estimation

b. Observations

c. Likelihood

d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: The probabilistic reasoning is used to represent uncertain knowledge, where we are not sure about the predicates. It depends Upon Estimation, Observation, and likelihood of objects.

---

44) The inference engine works on ______.

a. Forward Chaining

b. Backward Chaining

c. Both a and b

d. None of the above

Hide Answer Workspace

Answer: c. Both a and b

Explanation: The inference engine is the component of the intelligent system in artificial intelligence, which applies logical rules to the knowledge base to infer new information from known facts. The first inference engine was part of the expert system. Inference engine commonly proceeds in two modes, which are:

- **Forward chaining**
- **Backward chaining**

---

45) Which of the given statement is true for Conditional Probability?

a. Conditional Probability gives 100% accurate results.

b. Conditional Probability can be applied to a single event.

c. Conditional Probability has no effect or relevance on independent events.

d. None of the above.

Hide Answer Workspace

Answer: c. Conditional Probability has no effect or relevance on independent events.

Explanation: The conditional probability is said as the probability of occurring an event when another event has already occurred. And Independent events are those that are not affected by the occurrence of other events; hence conditional probability has no effect or relevance on independents events.

---

46) After applying conditional Probability to a given problem, we get______

a. 100% accurate result

b. Estimated Values

c. Wrong Values

d. None of the above

Hide Answer Workspace

Answer: b. Estimated Values

Explanation: Like all probability theories and methods, Conditional Probability also provides the estimated result value, which means the probability of an event to occur, not a 100% accurate result.

---

47) The best AI agent is one which___________

a. Needs user inputs for solving any problem
b. Can solve a problem on its own without any human intervention
c. Need a similar exemplary problem in its knowledge base
d. All of the above

Hide Answer Workspace

Answer: b. Can solve a problem on its own without any human intervention

Explanation: The best AI agent is one that can solve the problem on its own without any human intervention.

---

48) The Bayesian Network gives________

a. A complete description of the problem
b. Partial Description of the domain
c. A complete description of the domain
d. None of the above

Hide Answer Workspace

Answer: c. A complete description of the domain

Explanation: A Bayesian network

is a probabilistic graphical model that represents a set of variables and their conditional dependencies using a directed acyclic graph. It gives a complete description of the domain.

49) In LISP, the addition of 5+8 is entered as______.

a. 5+8
b. 5 add 8
c. 5+8=
d. (+5 8)

Hide Answer Workspace

Answer: d. (+5 8)

Explanation: The sum of two variables a & b can be entered as (+a b). Hence the sum of 5 and 8 can be entered as (+5 8).

---

50) An Algorithm is said as Complete algorithm if_____________

a. It ends with a solution (if any exists).
b. It begins with a solution.
c. It does not end with a solution.
d. It contains a loop

Hide Answer Workspace

Answer: a. It ends with a solution (if any exists).

Explanation: An algorithm is only said the complete algorithm if it ends with a solution (if it exists).

---

51) Which statement is valid for the Heuristic function?

a. The heuristic function is used to solve mathematical problems.
b. The heuristic function takes parameters of type string and returns an integer value.
c. The heuristic function does not have any return type.
d. The heuristic function calculates the cost of an optimal path between the pair of states.

Hide Answer Workspace

Answer: d. The heuristic function calculates the cost of an optimal path between the pair of states

Explanation: The heuristic function is used in Informed search in AI

to find the most promising path in the search. It estimates the closeness of the current state and calculates the cost of an optimal path between the pair of states. It is represented **by h(n).**

---

52) Which of the given element improve the performance of AI agent so that it can make better decisions?

a. Changing Element
b. Performance Element
c. Learning Element
d. None of the above

Hide Answer Workspace

Answer: c. Learning Element

Explanation: The learning element improves the performance of an AI agent while solving a given problem, so that it can make better decisions.

---

53) How many types of Machine Learning are there?

a. 1
b. 2
c. 3
d. 4

Hide Answer Workspace

Answer: c. 3

Explanation: There are three types of Machine Learning techniques, which are Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

54) The decision tree algorithm reaches its destination using____________.

a. Single Test

b. Two Test

c. Sequence of test

d. No test

Hide Answer Workspace

Answer: c. Sequence of test

Explanation: A decision tree is the supervised machine learning technique that can be used for both Classification and Regression problems. It reaches its destination using a Sequence of Tests.

---

55) In LISP programming, the square root is entered as_____.

a. Sqrt(x)

b. (sqrt x)

c. x/2

d. none of the above

Hide Answer Workspace

Answer: (sqrt x)

Explanation: In LISP programming, the square root of any variable x is entered as (sqrt x).

Artificial Intelligence Multiple Choice Questions

1) Artificial Intelligence is about_____.

a. Playing a game on Computer
b. Making a machine Intelligent
c. Programming on Machine with your Own Intelligence
d. Putting your intelligence in Machine

Hide Answer Workspace

Answer: b. Making a machine Intelligent.

Explanation: Artificial Intelligence is a branch of Computer science, which aims to create intelligent machines so that machine can think intelligently in the same manner as a human does.

---

2) Who is known as the -Father of AI"?

a. Fisher Ada
b. Alan Turing
c. John McCarthy
d. Allen Newell

Hide Answer Workspace

Answer: c. John McCarthy

Explanation: John McCarthy was a pioneer in the AI field and known as the father of Artificial intelligence. He was not only the known as the father of AI but also invented the term Artificial Intelligence.

---

3) Select the most appropriate situation for that a blind search can be used.

a. Real-life situation
b. Small Search Space
c. Complex game
d. All of the above

Hide Answer Workspace

Answer: b. Small Search Space

Explanation: Blind Search is also known as uninformed search, and it does not contain any domain information such as closeness, location of the goal, etc. Hence the most appropriate situation that can be used for the blind search is Small-search Space.

4) The application/applications of Artificial Intelligence is/are

a. Expert Systems
b. Gaming
c. Vision Systems
d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: All the given options are the applications of AI.

5) Among the given options, which search algorithm requires less memory?

a. Optimal Search
b. Depth First Search
c. Breadth-First Search
d. Linear Search

Hide Answer Workspace

Answer: b. Depth First Search

Explanation: The Depth Search Algorithm or DFS requires very little memory as it only stores the stack of nodes from the root node to the current node.

6) If a robot is able to change its own trajectory as per the external conditions, then the robot is considered as the__

a. Mobile
b. Non-Servo
c. Open Loop
d. Intelligent

Hide Answer Workspace

Answer: d. Intelligent

Explanation: If a robot is able to change its own trajectory as per the external conditions, then the robot is considered intelligent. Such type of agents come under the category of AI agents or Rational Agents.

---

7) Which of the given language is not commonly used for AI?

a. LISP
b. PROLOG
c. Python
d. Perl

Hide Answer Workspace

Answer: d. Perl

Explanation: Among the given languages, Perl is not commonly used for AI. LISP and PROLOG are the two languages that have been broadly used for AI innovation, and the most preferred language is Python for AI and Machine learning.

---

8) A technique that was developed to determine whether a machine could or could not demonstrate the artificial intelligence known as the___

a. Boolean Algebra
b. Turing Test
c. Logarithm

d. Algorithm

Hide Answer Workspace

Answer: b. Turing Test

Explanation: In the year **1950**, mathematician and computing pioneer **Alan Turing** introduced a test to determine whether a machine can think like a human or not, which means it can demonstrate intelligence, known as the **Turing Test**

. It was based on the **"Imitation game"** with some modifications. This technique is still a measure of various successful AI projects, with some updates.

---

9) The component of an Expert system is_________.

a. Knowledge Base
b. Inference Engine
c. User Interface
d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: Expert system is a part of AI and a computer program that is used to solve complex problems, and to give the decision-making ability like human. It does this with the help of **a Knowledge base, Inference engine, and User interface**, and all these are the components of an Expert System.

10) Which algorithm is used in the Game tree to make decisions of Win/Lose?

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b. DFS/BFS algorithm
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b. Discontinuous Variable
c. Both Discrete and Continuous variables
d. Continuous Variable only

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Answer: c. Both Discrete and Continuous Variables

Explanation: The Hybrid Bayesian network contains both discrete and continuous variables as the numerical inputs. To define the hybrid network, both kinds of distributions are used at wide probability distribution.

---

28) The process of capturing the inference process as Single Inference Rule is known as:

a. Clauses
b. Ponens
c. Generalized Modus Ponens
d. Variables

Hide Answer Workspace

Answer: c. Generalized Modus Ponens

Explanation: For all inference process in FOL, the single inference rule can be used, which is called Generalized Modus Ponens. It is said to be the lifted version of Modus ponens.

Generalized Modus Ponens can be said as, " **P implies Q and P is asserted to be true, therefore Q must be True."**

---

29) Which process makes two different Logical expressions look identical?

a. Unification
b. Lifting
c. Inference Process
d. None of the above

Hide Answer Workspace

Answer: a. Unification

Explanation: Unification is the process of making two different logical expressions identical by finding a substitution.

30) Which algorithm takes two sentences as input and returns a Unifier?

a. Inference
b. Hill-Climbing
c. Unify algorithm
d. Depth-first search

Hide Answer Workspace

Answer: c. Unify Algorithm

Explanation: The unify algorithm takes two atomic sentences and return a unifier. It is used for the unification process.

---

31) The PEAS in the task environment is about__________.

a. Peer, Environment, Actuators, Sense
b. Performance, Environment, Actuators, Sensors
c. Perceiving, Environment, Actuators, Sensors
d. None of the above

Hide Answer Workspace

Answer: b. Performance, Environment, Actuators, Sensors

Explanation: PEAS is a representation model on which an AI agent works. It is made up of four words:

- P: Performance
- E: Environment
- A: Actuators
- S: Sensors

---

32) In state-space, the set of actions for a given problem is expressed by the_____.

a. Intermediate States

b. Successor function that takes current action and returns next state

c. Initial States

d. None of the above

Hide Answer Workspace

Answer: b. Successor function that takes current action and returns next state

Explanation: The successor function provides a description of all possible actions and their next states, which means their outcomes.

---

33) In which search problem, to find the shortest path, each city must be visited once only?

a. Map coloring Problem

b. Depth-first search traversal on a given map represented as a graph

c. Finding the shortest path between a source and a destination

d. Travelling Salesman problem

Hide Answer Workspace

Answer: d. Travelling Salesman problem

Explanation: The TSP or Travelling Salesman problem is about finding the shortest possible route to visit each city only once and returning to the origin city when the list of all cities and distances between each pair of cities is given.

---

34) In the TSP problem of n cities, the time taken for traversing all cities, without having prior knowledge of the length of the minimum tour will be_______.

a. O(n)

b. O(n2)

c. O(n!)

d. O(n/2)

Hide Answer Workspace

Answer: c. O(n!)

Explanation: In the TSP problem of n cities, the time taken for traversing all cities without having prior knowledge of the length of the minimum tour will be O(n!).

---

35) Web Crawler is an example of______.

a. Intelligent Agent
b. Problem-solving agent
c. Simple reflex agent
d. Model-based agent

Hide Answer Workspace

Answer: Intelligent Agent

Explanation: The web crawler is an example of Intelligent agents, which is responsible for collecting resources from the Web, such as HTML documents, images, text files, etc.

---

36) The main function of problem-solving agent is to________.

a. Solve the given problem and reach the goal
b. Find out which sequence of action will get it to the goal state.
c. Both a & b
d. None of the above

Hide Answer Workspace

Answer: Both a & b

Explanation: Problem-solving agents are the goal-based agents that use different search strategies and algorithms to solve a given problem.

---

37) In artificial Intelligence, knowledge can be represented as_______.

i. Predicate Logic

ii. Propositional Logic

iii. Compound Logic

iv. Machine Logic

a. Both I and II
b. Only II
c. Both II and III
d. Only IV

Hide Answer Workspace

Answer: a. Both I and II

Explanation: There are several techniques of Knowledge representation in AI,

and among them, one is Logical Representation. The logical representation can be done in two ways **Predicate Logic and Propositional Logic**, hence knowledge can be represented as both predicate and Propositional logic.

---

38) For propositional Logic, which statement is false?

a. The sentences of Propositional logic can have answers other than True or False.
b. Each sentence is a declarative sentence.
c. Propositional logic is a knowledge representation technique in AI.
d. None of the above

Hide Answer Workspace

Answer: a. The sentences of Propositional logic can have answers other than True or False

Explanation: Propositional Knowledge or PL is the simplest form of logic that is used to represent the knowledge, where all the sentences are propositions. In this, each sentence is a declarative sentence that can only be either true or False.

Such as, **It is Sunday today.** This sentence can be either true or false only.

---

39) First order logic Statements contains______.

a. Predicate and Preposition
b. Subject and an Object
c. Predicate and Subject
d. None of the above

Hide Answer Workspace

Answer: c. Predicate and Subject

Explanation: The first-order logic

is also known as the First-order predicate logic, which is another way of knowledge representation. The FOL statements contain two parts that are s**ubject and Predicate**.

For e.g., **X is an Integer; In this, X is Subject and Is an Integer is Predicate**.

40) A knowledge-based agent can be defined with _____ levels.

a. 2 Levels
b. 3 Levels
c. 4 Levels
d. None of the above

Hide Answer Workspace

Answer: b. 3 Levels

Explanation: The knowledge-based agents have the capability of making decisions and reasoning to act efficiently. It can be viewed at three different levels, which are:

- **Knowledge Level**
- **Logical Level**
- **Implementation Level**

---

41) Ways to achieve AI in real-life are_________.

a. Machine Learning

b. Deep Learning

c. Both a & b

d. None of the above

Hide Answer Workspace

Answer: c. Both a &b

Explanation: Machine Learning and Deep Learning are the two ways to achieve AI in real life.

---

42) The main tasks of an AI agent are_______.

a. Input and Output

b. Moment and Humanly Actions

c. Perceiving, thinking, and acting on the environment

d. None of the above

Hide Answer Workspace

Answer: c. Perceiving, thinking, and acting on the environment

Explanation: The AI agent is the rational agent that runs in the cycle of Perceive, think, and act.

---

43) The probabilistic reasoning depends upon____________.

a. Estimation

b. Observations

c. Likelihood

d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: The probabilistic reasoning is used to represent uncertain knowledge, where we are not sure about the predicates. It depends Upon Estimation, Observation, and likelihood of objects.

---

44) The inference engine works on ______.

a. Forward Chaining
b. Backward Chaining
c. Both a and b
d. None of the above

Hide Answer Workspace

Answer: c. Both a and b

Explanation: The inference engine is the component of the intelligent system in artificial intelligence, which applies logical rules to the knowledge base to infer new information from known facts. The first inference engine was part of the expert system. Inference engine commonly proceeds in two modes, which are:

- **Forward chaining**
- **Backward chaining**

---

45) Which of the given statement is true for Conditional Probability?

a. Conditional Probability gives 100% accurate results.
b. Conditional Probability can be applied to a single event.
c. Conditional Probability has no effect or relevance on independent events.
d. None of the above.

Hide Answer Workspace

Answer: c. Conditional Probability has no effect or relevance on independent events.

Explanation: The conditional probability is said as the probability of occurring an event when another event has already occurred. And Independent events are those that are not affected by the occurrence of other events; hence conditional probability has no effect or relevance on independents events.

---

46) After applying conditional Probability to a given problem, we get______

a. 100% accurate result
b. Estimated Values
c. Wrong Values
d. None of the above

Hide Answer Workspace

Answer: b. Estimated Values

Explanation: Like all probability theories and methods, Conditional Probability also provides the estimated result value, which means the probability of an event to occur, not a 100% accurate result.

---

47) The best AI agent is one which____________

a. Needs user inputs for solving any problem
b. Can solve a problem on its own without any human intervention
c. Need a similar exemplary problem in its knowledge base
d. All of the above

Hide Answer Workspace

Answer: b. Can solve a problem on its own without any human intervention

Explanation: The best AI agent is one that can solve the problem on its own without any human intervention.

---

48) The Bayesian Network gives________

a. A complete description of the problem
b. Partial Description of the domain

c. A complete description of the domain
d. None of the above

Hide Answer Workspace

Answer: c. A complete description of the domain

Explanation: A Bayesian network

is a probabilistic graphical model that represents a set of variables and their conditional dependencies using a directed acyclic graph. It gives a complete description of the domain.

---

49) In LISP, the addition of 5+8 is entered as______.

a. 5+8
b. 5 add 8
c. 5+8=
d. (+5 8)

Hide Answer Workspace

Answer: d. (+5 8)

Explanation: The sum of two variables a & b can be entered as (+a b). Hence the sum of 5 and 8 can be entered as (+5 8).

---

50) An Algorithm is said as Complete algorithm if_____________

a. It ends with a solution (if any exists).
b. It begins with a solution.
c. It does not end with a solution.
d. It contains a loop

Hide Answer Workspace

Answer: a. It ends with a solution (if any exists).

Explanation: An algorithm is only said the complete algorithm if it ends with a solution (if it exists).

---

51) Which statement is valid for the Heuristic function?

a. The heuristic function is used to solve mathematical problems.
b. The heuristic function takes parameters of type string and returns an integer value.
c. The heuristic function does not have any return type.
d. The heuristic function calculates the cost of an optimal path between the pair of states.

Hide Answer Workspace

Answer: d. The heuristic function calculates the cost of an optimal path between the pair of states

Explanation: The heuristic function is used in Informed search in AI

to find the most promising path in the search. It estimates the closeness of the current state and calculates the cost of an optimal path between the pair of states. It is represented **by h(n).**

---

52) Which of the given element improve the performance of AI agent so that it can make better decisions?

a. Changing Element
b. Performance Element
c. Learning Element
d. None of the above

Hide Answer Workspace

Answer: c. Learning Element

Explanation: The learning element improves the performance of an AI agent while solving a given problem, so that it can make better decisions.

53) How many types of Machine Learning are there?

a. 1
b. 2
c. 3
d. 4

Hide Answer Workspace

Answer: c. 3

Explanation: There are three types of Machine Learning techniques, which are Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

54) The decision tree algorithm reaches its destination using_____________.

a. Single Test
b. Two Test
c. Sequence of test
d. No test

Hide Answer Workspace

Answer: c. Sequence of test

Explanation: A decision tree is the supervised machine learning technique that can be used for both Classification and Regression problems. It reaches its destination using a Sequence of Tests.

55) In LISP programming, the square root is entered as_____.

a. Sqrt(x)
b. (sqrt x)
c. x/2
d. none of the above

Hide Answer Workspace

Answer: (sqrt x)

Explanation: In LISP programming, the square root of any variable x is entered as (sqrt x).

Remember In Prayers

Good Luck